WRITE BOTH ALGORITHM AND PYTHON PROGRAM WITH OUTPUT

1. **Sum of Two Numbers**
    - **Problem**: Input two numbers and display their sum.
    - **Concepts**: Input/Output, Arithmetic
2. **Find Area of Rectangle**
    - **Problem**: Input length and breadth and calculate the area.
    - **Formula**: `Area = length × breadth`
3. **Convert Celsius to Fahrenheit**
    - **Formula**: `F = (C × 9/5) + 32`
    - **Problem**: Input temperature in Celsius and convert it to Fahrenheit.
4. **Simple Interest Calculator**
    - **Formula**: `SI = (P × R × T) / 100`
    - **Input**: Principal, Rate, Time
5. **Check Whether Number is Positive or Negative**
    - **Concept**: Use of decision (if-else) in algorithm and flowchart.

🔸 Medium Level (Conditions and Loops)

6. **Find the Greater of Two Numbers**
    - **Input**: Two numbers
    - **Output**: Which number is greater
7. **Even or Odd Checker**
    - **Input**: One number
    - **Logic**: Check if the number is divisible by 2

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8. **Check for Leap Year**
    - **Condition**:
        - Divisible by 4 and not by 100
        - Or divisible by 400

9. **Print Numbers from 1 to 10**

    - **Concept**: Looping (repetition) in flowchart

10. **Calculate Factorial of a Number**

    - **Input**: A number `n`

    - **Output**: Factorial of `n`

    - **Formula**: `n! = n × (n-1) × ... × 1`

11. Print all even numbers between 1 and 50.

12. Print the multiplication table of a number entered by the user.

13. Print the sum of the first 10 natural numbers.

14. Print the square of numbers from 1 to 10.

15. Count how many vowels are in the word `"Education"` using a for loop.

16. Print all numbers from 1 to 100 that are divisible by 3 and 5.

17. Print the factorial of a number entered by the user using a for loop.

18. Print each character of a string `"Python is fun"` using a for loop.

19. Print the reverse of a string using a for loop

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🔺 Creative / Word Problems (Challenge)

20. **Electricity Bill Calculator**
    - **Condition**:
        - First 100 units: ₹5/unit
        - Next 100 units: ₹7/unit
        - Above 200: ₹10/unit
21. **Grade Calculator**
    - **Input**: Marks in 5 subjects
    - **Logic**: Calculate average and assign grade
        - 90+: A
        - 80–89: B
        - 70–79: C
        - Below 70: D
22. **ATM Withdrawal Simulation**
    - Input: Amount
    - Output: Number of notes (100s, 500s, etc.)
23. **Check if Triangle is Valid**
    - Input: 3 angles
    - Condition: If sum = 180 → Valid triangle
24. **Simple Voting Eligibility**
    - Input: Age
    - Output: Eligible to vote or not (age ≥ 18)